

W6: Many-Body Perturbation Theory and Feynman Diagrams

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Section 9.1: The Coulomb Interaction and Perturbation Theory – Objective

The aim is to calculate material properties via Green's functions by treating the Coulomb interaction as a perturbation. Since the non-interacting problem can be solved exactly, the strategy is to expand around that solution.

Scattering Theory: Lippmann–Schwinger Equation

In scattering theory, an unperturbed state $|\phi_0\rangle$ is modified by a perturbation $\hat{V}$. The Lippmann–Schwinger equation is:

$$|\phi\rangle = |\phi_0\rangle + \frac{1}{\omega - \hat{H}_0 + i\eta} \hat{V} |\phi\rangle. \quad (9.1)$$

This can be rewritten using the non-interacting Green's function

$$G_0 \equiv \frac{1}{\omega + i\eta - \hat{H}_0},$$

yielding:

$$|\phi\rangle = \left(1 - G_0 \hat{V}\right)^{-1} |\phi_0\rangle. \quad (9.2)$$

Connecting Scattering Theory to Perturbation Theory

The expansion

$$\left(1 - G_0 \hat{V}\right)^{-1} = 1 + G_0 \hat{V} + G_0 \hat{V} G_0 \hat{V} + \cdots ,$$

is the perturbative (Born) series.

Each term in the series corresponds to successive scattering events.

This series forms the basis for constructing the full Green's function.

From Scattering to the Dyson Equation

The full Green's function is defined by:

$$G(\omega) \equiv \frac{1}{\omega + i\eta - \hat{H}},$$

where $\hat{H} = \hat{H}_0 + \hat{V}$.

Using the series expansion from scattering theory, we have:

$$G = G_0 + G_0 \hat{V} G_0 + G_0 \hat{V} G_0 \hat{V} G_0 + \dots$$

This expansion is compactly written as the Dyson equation:

$$G = G_0 + G_0 \Sigma G, \tag{9.3}$$

where the self-energy Σ summarizes all scattering (interaction) processes.

Scattering in the Continuum and Green's Functions

In the continuum, the eigenstates are extended (scattering states) rather than bound.

Physical observables (e.g. scattering amplitudes) depend on the **asymptotic behavior** of these states.

The Green's function, with its $i\eta$ prescription, encodes the causal propagation of scattering states.

Thus, in the continuum limit, we only need the asymptotic (scattered) part of the wavefunction to compute observables.

Continuum Eigenstates: Single-Particle vs. Many-Body

In scattering theory, energies in the continuum correspond to extended states. For a non-interacting Hamiltonian, these eigenstates are typically single-particle states (e.g. plane waves) that are not normalizable.

In a full many-body system, the eigenstates are many-body states (often combinations of many Slater determinants) and are also extended.

When working with one-particle Green's functions, we usually consider transitions where an electron is added or removed. In this effective picture, the relevant excitations are often treated as single-particle (or quasi-particle) states.

Thus, while the full spectrum of the many-body Hamiltonian is built from many-body eigenstates, in practical scattering and Green's function calculations we often focus on the effective single-particle excitations that live in the continuum.

Section 9.1: Perturbation Theory Approaches

Brillouin–Wigner Perturbation Theory:

$$E = E_{ip} + \sum_{n=1}^{\infty} \langle \phi_{ip} | \hat{V} \left(\frac{1}{1 - \hat{P}} \frac{1}{E - \hat{H}_0} \hat{V} \right)^{n-1} | \phi_{ip} \rangle, \quad (9.5)$$

and

$$|\Phi\rangle = \sum_{n=0}^{\infty} \left(\frac{1}{1 - \hat{P}} \frac{1}{E - \hat{H}_0} \hat{V} \right)^n | \phi_{ip} \rangle. \quad (9.6)$$

Rayleigh–Schrödinger Perturbation Theory:

$$E = E_{ip} + \sum_{n=1}^{\infty} \langle \phi_{ip} | \hat{V} \left(\frac{1}{E_{ip} - \hat{H}_0} (E_{ip} - E + \hat{V}) \right)^{n-1} | \phi_{ip} \rangle, \quad (9.7)$$

and

$$|\Phi\rangle = \sum_{n=0}^{\infty} \left(\frac{1}{E_{ip} - \hat{H}_0} (E_{ip} - E + \hat{V}) \right)^n | \phi_{ip} \rangle. \quad (9.8)$$

Section 9.1: Møller–Plesset Perturbation Theory

Instead of treating the full Coulomb interaction as a perturbation, a part is incorporated into a mean-field Hamiltonian.

Zeroth-order energy: Sum of occupied HF eigenvalues.

First-order correction is

$$E^{(1)} = \langle \phi_{ip} | \hat{V} | \phi_{ip} \rangle = - \sum_{i < j} (v_{ijji} - v_{ijij}), \quad (9.9)$$

so that $E_{ip} + E^{(1)}$ equals the HF total energy.

Second-order correction (MP2):

$$E^{(2)} = \sum_{i < j}^{\text{occ}} \sum_{a < b}^{\text{empty}} \frac{|v_{ijab} - v_{ijba}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}. \quad (9.11)$$

Section 9.2: Objective

This section addresses the difficulty that the interacting ground state appears explicitly in the Green's function expressions. The strategy is to connect the non-interacting and interacting systems via **adiabatic switching**.

Adiabatic Switching-On of the Interaction

The Coulomb interaction is slowly switched on using a time-dependent form:

$$v_{\eta}(t) = v_c e^{-\eta|t|}, \quad \eta \rightarrow 0^+.$$

At $t \rightarrow \pm\infty$, the interaction is off; at $t = 0$, the full interaction is present.

Time Evolution Operator in the Interaction Picture

In the interaction picture, the full Hamiltonian is split as:

$$H = H_0 + V,$$

where H_0 is the non-interacting (free) part and V is the interaction.

The time evolution operator $U_I(t, t_0)$ evolves states according to the interaction:

$$|\psi_I(t)\rangle = U_I(t, t_0) |\psi_I(t_0)\rangle.$$

$U_I(t, t_0)$ satisfies the differential equation:

$$\frac{d}{dt} U_I(t, t_0) = -i V_I(t) U_I(t, t_0), \quad U_I(t_0, t_0) = I,$$

where the interaction-picture operator is given by:

$$V_I(t) = e^{iH_0(t-t_0)} V e^{-iH_0(t-t_0)}.$$

The formal solution is expressed as a time-ordered exponential:

$$U_I(t, t_0) = T \left\{ \exp \left[-i \int_{t_0}^t V_I(t') dt' \right] \right\}.$$

Understanding the Time-Ordered Series

The time evolution operator in the interaction picture is defined as

$$U_I(t, t_0) = T \left\{ \exp \left[-i \int_{t_0}^t V_I(t') dt' \right] \right\},$$

where T denotes time ordering.

The role of T is to rearrange operators so that those with later times appear to the left:

$$T \{ V_I(t_1) V_I(t_2) \} = \begin{cases} V_I(t_1) V_I(t_2), & t_1 > t_2, \\ V_I(t_2) V_I(t_1), & t_2 > t_1. \end{cases}$$

Expanding the exponential yields:

$$U_I(t, t_0) = 1 + (-i) \int_{t_0}^t dt_1 V_I(t_1) + \frac{(-i)^2}{2!} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 T \{ V_I(t_1) V_I(t_2) \} + \dots$$

Equivalently, by using ordered integration limits we can write:

$$U_I(t, t_0) = 1 + (-i) \int_{t_0}^t dt_1 V_I(t_1) + (-i)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 V_I(t_1) V_I(t_2) + \dots$$

Each term is “sorted in orders of V ” – the n th term is proportional to V^n .

Propagating the Ground State In the interaction picture

In the interaction picture, we propagate the non-interacting ground state $|\phi_{ip}\rangle$ from $t = -\infty$ to time t and further to $t = \infty$.

The time evolution operator $U_I(t, t_0)$ connects the non-interacting state to the interacting state.

This gives the relation:

$$\langle \phi_{\eta}^+ | \hat{O}_H(t) | \phi_{\eta}^- \rangle = \langle \phi_{ip} | U_I(\infty, t) \hat{O}_I(t) U_I(t, -\infty) | \phi_{ip} \rangle.$$

Here, $\hat{O}_H(t)$ is the operator in the Heisenberg picture and $\hat{O}_I(t)$ in the interaction picture.

Normalization and Removal of Divergent Phases

Direct propagation introduces divergent phases due to adiabatic switching.

To cancel these divergences, we normalize the expectation value by the overlap of the evolved states.

The normalized expression is:

$$\frac{\langle \phi_{\eta}^+ | \hat{O}_I(t) | \phi_{\eta}^- \rangle}{\langle \phi_{\eta}^+ | \phi_{\eta}^- \rangle} = \frac{\langle \phi_{ip} | U_I(\infty, t) \hat{O}_I(t) U_I(t, -\infty) | \phi_{ip} \rangle}{\langle \phi_{ip} | U_I(\infty, -\infty) | \phi_{ip} \rangle}.$$

This ratio ensures that the divergent phase factors cancel, leaving a well-defined expression.

Expansion of the Expectation Value

The time evolution operator U_I can be expanded in a Dyson series, order by order in the interaction V_I .

The expansion for the expectation value of an operator $\hat{O}_H(t)$ becomes:

$$\begin{aligned} \lim_{\eta \rightarrow 0} \langle \phi_\eta^+ | \hat{O}_H(t) | \phi_\eta^- \rangle &= \lim_{\eta \rightarrow 0} \sum_{p=0}^{\infty} \frac{(-i)^p}{p!} \int_{-\infty}^{\infty} dt_1 \cdots \int_{-\infty}^{\infty} dt_p e^{-\eta(|t_1| + \cdots + |t_p|)} \\ &\quad \times \langle \phi_{ip} | T \{ V_I(t_1) \cdots V_I(t_p) \hat{O}_I(t) \} | \phi_{ip} \rangle \langle \phi_{ip} | S_\eta | \phi_{ip} \rangle. \end{aligned} \quad (9.20)$$

Here:

T denotes time ordering.

$S_\eta = U_I(\infty, -\infty)$ is the full time-evolution (scattering) operator.

The exponential factors $e^{-\eta|t_i|}$ implement adiabatic switching.

This expansion forms the basis for the diagrammatic representation of the interacting Green's function.

Connecting Ground States and the Gell-Mann–Low Theorem

Propagate the non-interacting ground state $|\phi_{ip}\rangle$ from $t = -\infty$ to $t = 0$ using U_I .

Define the states $|\phi_\eta^-(t)\rangle$ and $|\phi_\eta^+(t)\rangle$ by forward and backward propagation, respectively.

In the limit $\eta \rightarrow 0$, these states converge to the true interacting ground state $|\phi_0\rangle$ (This is the main hypothesis of adiabatic switching).

Expectation values are computed as:

$$\langle O \rangle = \frac{\langle \phi_\eta^+ | O | \phi_\eta^- \rangle}{\langle \phi_\eta^+ | \phi_\eta^- \rangle}.$$

This is the Gell-Mann–Low theorem guarantees that adiabatic switching connects the non-interacting ground state to an interacting state (No necessarily the ground state!!, although we assume so.).

Interacting Green's Function

The zero-temperature interacting Green's function is:

$$G(x, t; x', t') = -i \frac{\langle \phi_{ip} | T \{ \psi_I(x, t) \psi_I^\dagger(x', t') S_\eta \} | \phi_{ip} \rangle}{\langle \phi_{ip} | S_\eta | \phi_{ip} \rangle},$$

where

$$S_\eta \equiv U_I(\infty, -\infty).$$

Diagrammatic Expansion of the Green's Function

The interacting Green's function at zero temperature is given by:

$$G(x, t; x', t') = -i \frac{\langle \phi_{ip} | T \left[\psi(x, t) \psi^\dagger(x', t') \hat{S} \right] | \phi_{ip} \rangle}{\langle \phi_{ip} | \hat{S} | \phi_{ip} \rangle}, \quad (9.22)$$

where $\hat{S}_\eta = U_I(\infty, -\infty)$.

The Coulomb interaction in the interaction picture is:

$$\hat{V}_I(t) = \frac{1}{2} \int dx dx' \psi_I^\dagger(x, t) \psi_I^\dagger(x', t) v_c(r, r') \psi_I(x', t) \psi_I(x, t). \quad (9.23)$$

Expanding the numerator yields a series in powers of v_c .

Compact Notation for the Diagrammatic Expansion

In compact notation, the expansion becomes (schematically):

$$\begin{aligned} -i \langle \phi_{ip} | T \left[\psi(x, t) \psi^\dagger(x', t') \hat{S} \right] | \phi_{ip} \rangle &= G_0(x, t; x', t') \\ &+ \sum_{p=1}^{\infty} \left(\frac{i}{2} \right)^p \frac{1}{p!} \int \cdots \int d1 d1' \cdots dp dp' v_c(1, 1') \cdots v_c(p, p') \\ &\times G_{2p+1}^0(x, t, 1, 1', \dots, p, p'; x', t', 1, 1', \dots, p, p'). \end{aligned}$$

Here, G_s^0 denotes the non-interacting Green's function involving s pairs of field operators (as defined in Eq. (9.26)).

Understanding the Notation $G_{(2p+1)}^0$

The one-body Green's function is defined as

$$G^0(x, t; x', t') = \langle \phi_{ip} | T \left[\psi_I(x, t) \psi_I^\dagger(x', t') \right] | \phi_{ip} \rangle,$$

which we can denote as $G_0^{(1)}$ (i.e. one pair of field operators or one propagator line).

In our perturbative expansion (Eq. (9.24)), each Coulomb vertex (a two-body operator) effectively adds one extra pair of field operators.

Therefore, when p vertices are inserted:

$$\text{Total number of operator pairs} = 1 + 2p = 2p + 1,$$

meaning that the corresponding Green's function involves $2p + 1$ propagator lines (or equivalently, $2(2p + 1)$ individual field operators).

Thus, the notation $G_{(2p+1)}^0$ denotes the non-interacting Green's function constructed from a time-ordered product corresponding to $2p + 1$ propagator lines. Here, p is the order of the perturbation (i.e. the number of Coulomb interaction insertions).

Recap and Physical Interpretation

The expansion in Eq. (9.24) reads schematically as:

$$\begin{aligned} -i \langle \phi_{ip} | T [\psi(x, t) \psi^\dagger(x', t') \hat{S}] | \phi_{ip} \rangle &= G^0(x, t; x', t') \\ &+ \sum_{p=1}^{\infty} \left(\frac{i}{2} \right)^p \frac{1}{p!} \int \cdots \int d1 d1' \cdots dp dp' v_c(1, 1') \cdots v_c(p, p') \\ &\times G_{(2p+1)}^0(x, t, 1, 1', \dots, p, p'; x', t', 1, 1', \dots, p, p'). \end{aligned}$$

The subscript $2p + 1$ in $G_{(2p+1)}^0$ indicates that, after inserting p Coulomb vertices, The original two external operators are supplemented by $2p$ operators from the vertices.

This results in a Green's function containing a total of $2p + 1$ propagator lines. In this context, p is indeed the order of the perturbation, and the notation reflects how the diagrammatic structure grows with each interaction insertion.

Interpreting the G_0^3 Term in First-Order Perturbation

In the expansion of the interacting Green's function, first-order perturbation involves one Coulomb vertex.

Expanding the time-evolution operator (via the Dyson series) and contracting the field operators with Wick's theorem yields a term proportional to G_0^3 .

This G_0^3 term involves a time-ordered product of **six** field operators (three creation and three annihilation operators), which is the definition of the three-body non-interacting Green's function.

Physical interpretation: The extra pair of field operators arises from the Coulomb interaction, which creates (and later annihilates) an electron-hole pair. Thus, the diagram includes:

The propagation of the original (external) electron,

Plus an additional electron-hole excitation (represented by two extra G_0 lines).

In this sense, one might loosely say it “describes the propagation of three particles,” but more precisely it represents the additional propagation channels (i.e. the extra electron-hole pair) induced by the interaction.

Higher-Order Non-interacting Green's Functions and Their Determinant Form

The one-body non-interacting Green's function is defined as

$$G_0(x, t; x', t') = \langle \phi_{ip} | T \left[\psi_I(x, t) \psi_I^\dagger(x', t') \right] | \phi_{ip} \rangle. \quad (9.25)$$

The s -particle non-interacting Green's function is denoted by G_s^0 and defined by

$$G_s^0(1, 2, \dots, s; 1', 2', \dots, s') = (-i)^s \langle \phi_{ip} | T \left[\psi_I(1) \psi_I(2) \cdots \psi_I(s) \psi_I^\dagger(s') \cdots \psi_I^\dagger(1') \right] | \phi_{ip} \rangle. \quad (9.26)$$

Here, the subscript s indicates that the Green's function involves s pairs of field operators (or s particles).

Example: (Problem 9.2) The two-particle Green's function is

$$G_2^0(1, 2; 1', 2') = G_0(1, 1') G_0(2, 2') - G_0(1, 2') G_0(2, 1') = \begin{vmatrix} G_0(1, 1') & G_0(1, 2') \\ G_0(2, 1') & G_0(2, 2') \end{vmatrix}. \quad (9.27)$$

Problem Statement

Goal: Calculate the two-particle non-interacting Green's function:

$$G_2^0(1, 2; 1', 2') = (-i)^2 \langle \phi_{ip} | T \{ \psi_I(1) \psi_I(2) \psi_I^\dagger(2') \psi_I^\dagger(1') \} | \phi_{ip} \rangle.$$

Here, 1 denotes the set (x_1, t_1) and similarly for 2, 1', 2'.

$|\phi_{ip}\rangle$ is the non-interacting ground state (a single Slater determinant).

Direct Evaluation Using Slater–Condon Rules

In a single Slater determinant, the many-body wave function is given by a determinant of single-particle orbitals.

The expectation value of a product of field operators over a Slater determinant decomposes into sums over products of one-body matrix elements.

For the product

$$T\{\psi_I(1)\psi_I(2)\psi_I^\dagger(2')\psi_I^\dagger(1')\},$$

nonzero contributions arise only when every annihilation operator finds a corresponding creation operator.

There are two possible pairings:

- (a) Pair $\psi_I(1)$ with $\psi_I^\dagger(1')$ and $\psi_I(2)$ with $\psi_I^\dagger(2')$.
- (b) Pair $\psi_I(1)$ with $\psi_I^\dagger(2')$ and $\psi_I(2)$ with $\psi_I^\dagger(1')$.

Final Expression for G_2^0

Define the one-body non-interacting Green's function as:

$$G_0(1, 1') = \langle \phi_{ip} | T \{ \psi_I(1) \psi_I^\dagger(1') \} | \phi_{ip} \rangle.$$

Then, by the Slater–Condon rules, the two possible pairings yield:

(a) $G_0(1, 1') G_0(2, 2')$

(b) $G_0(1, 2') G_0(2, 1')$

Due to the antisymmetry of the fermionic wave function, the second pairing acquires a minus sign.

Therefore, we have:

$$G_2^0(1, 2; 1', 2') = (-i)^2 \left[G_0(1, 1') G_0(2, 2') - G_0(1, 2') G_0(2, 1') \right].$$

Since $(-i)^2 = -1$ (with the overall sign convention adjusted), the final expression is commonly written as:

$$G_2^0(1, 2; 1', 2') = G_0(1, 1') G_0(2, 2') - G_0(1, 2') G_0(2, 1').$$

Diagrammatic Representation of the Expansion

Diagrams represent the expansion:

Zeroth order: A single propagator G_0 .

First order: Diagrams with one Coulomb interaction (Hartree and Fock diagrams).

Higher orders: Diagrams with multiple vertices.

Diagrams are classified as:

Connected: Cannot be separated into parts by cutting one propagator.

Disconnected: Represent vacuum fluctuations; cancel out via the linked cluster theorem.

Wick's Theorem

Wick's theorem expresses time-ordered products of field operators as sums over all possible contractions.

For two operators:

$$\dot{A}\dot{B} = \hat{A}\hat{B} - N(\hat{A}\hat{B}),$$

where the contraction is:

$$\dot{A}\dot{B} = \langle 0 | \hat{A}\hat{B} | 0 \rangle.$$

For many operators:

$$\hat{A}\hat{B}\hat{C} \cdots \hat{X}\hat{Y}\hat{Z} = N(\hat{A}\hat{B} \cdots \hat{Z}) + N(\text{all possible contractions}).$$

(See Eq. 9.32)

This leads to the determinant structure of higher-order Green's functions.

Linked Cluster Theorem

The linked cluster theorem guarantees that disconnected (vacuum) diagrams cancel out when the Green's function is properly normalized.

Although the expansion contains many disconnected diagrams, they appear in both the numerator and denominator and cancel.

Thus, only the connected diagrams contribute to the physical Green's function.

Conclusions

The expansion of the Green's function in many-body perturbation theory can be organized diagrammatically.

Wick's theorem simplifies the evaluation of time-ordered products, leading to determinant expressions.

The linked cluster theorem ensures that only connected diagrams contribute to physical observables.

This framework sets the stage for practical approximations (e.g., Dyson equation, self-energy functionals) in later chapters.

Appendix I: Time Evolution in the Interaction Picture

Split the full Hamiltonian:

$$H = H_0 + V.$$

Operator evolution:

$$O_I(t) = e^{iH_0(t-t_0)} O_S e^{-iH_0(t-t_0)}.$$

State evolution:

$$|\psi_I(t)\rangle = U_I(t, t_0) |\psi_I(t_0)\rangle,$$

with

$$U_I(t, t_0) = \mathcal{T} \left\{ \exp \left[-i \int_{t_0}^t V_I(t') dt' \right] \right\}.$$

Appendix II: Feynman Diagrams in Many-Body Theory – Basics

Propagators: The bare propagator $G_0(\omega)$ represents the amplitude for a particle to propagate:

$$G_0(\omega) = \frac{1}{\omega + i\eta - \varepsilon_k}.$$

Interaction Vertices: The Coulomb interaction:

$$v_c(r - r') = \frac{e^2}{|r - r'|},$$

is depicted by dashed lines.

Appendix II: Feynman Diagrams – Rules and Expansion

Feynman Rules:

- (a) Assign a propagator G_0 to each line.
- (b) Each interaction vertex contributes a factor of v_c and an integration over space-time.
- (c) Include symmetry factors and minus signs (for fermions).

Diagrammatic Expansion: The full Green's function is expanded as:

$$G = G_0 + G_0 \hat{V} G_0 + G_0 \hat{V} G_0 \hat{V} G_0 + \cdots .$$

Appendix II: Single & Double Excitations and Brillouin's Theorem

Single Excitations: Promoting one electron from an occupied to an unoccupied orbital. Brillouin's theorem tells us:

$$\langle \Phi_{\text{HF}} | \hat{H} | \Phi_i^a \rangle = 0.$$

Double Excitations: Simultaneous excitation of two electrons. These contribute at second order.

Appendix II: Vacuum Fluctuations and the Linked Cluster Theorem

Disconnected (vacuum) diagrams appear in the expansion.

The linked cluster theorem guarantees that these cancel in the normalized Green's function.

Appendix III: Overview of the Slater–Condon Rules

The Slater–Condon rules provide a systematic way to evaluate matrix elements of one-body and two-body operators between Slater determinants.

A **Slater determinant** is an antisymmetrized product of single-particle orbitals representing a many-electron wave function.

These rules are essential in quantum chemistry and many-body physics for simplifying calculations in methods like Hartree–Fock and configuration interaction.

One-Body Operators

A one-body operator, $\hat{O}^{(1)}$, (e.g. kinetic energy or an external potential) acts on a single electron.

Diagonal matrix elements:

$$\langle \Phi | \hat{O}^{(1)} | \Phi \rangle = \sum_{i \in \text{occ}} \langle \phi_i | \hat{o}^{(1)} | \phi_i \rangle.$$

Off-diagonal matrix elements: Nonzero only if the two Slater determinants differ by one orbital. The matrix element is given by the single-particle integral:

$$\langle \Phi_i^a | \hat{O}^{(1)} | \Phi \rangle = \langle \phi_a | \hat{o}^{(1)} | \phi_i \rangle,$$

where $|\Phi_i^a\rangle$ is obtained by replacing the occupied orbital ϕ_i with a virtual orbital ϕ_a .

Two-Body Operators

A two-body operator, $\hat{O}^{(2)}$, (e.g. the Coulomb interaction) acts on pairs of electrons.

Diagonal matrix elements:

$$\langle \Phi | \hat{O}^{(2)} | \Phi \rangle = \frac{1}{2} \sum_{i,j \in \text{occ}} \left(\langle \phi_i \phi_j | \hat{o}^{(2)} | \phi_i \phi_j \rangle - \langle \phi_i \phi_j | \hat{o}^{(2)} | \phi_j \phi_i \rangle \right).$$

Off-diagonal matrix elements: Nonzero only if the determinants differ by at most two orbitals.

For determinants that differ by two orbitals (say, ϕ_i, ϕ_j replaced by ϕ_a, ϕ_b):

$$\langle \Phi_{ij}^{ab} | \hat{O}^{(2)} | \Phi \rangle = \langle \phi_a \phi_b | \hat{o}^{(2)} | \phi_i \phi_j \rangle - \langle \phi_a \phi_b | \hat{o}^{(2)} | \phi_j \phi_i \rangle.$$

If the determinants differ by more than two orbitals, the matrix element is zero.